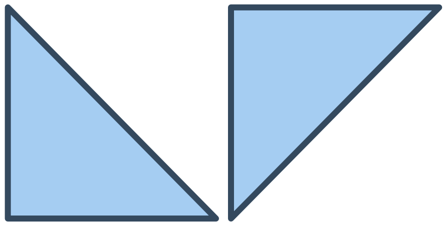
Translations

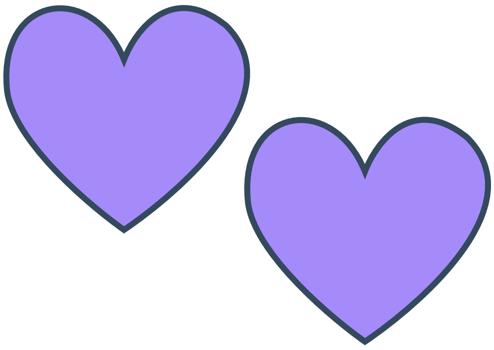
1 State the type of transformattion that has occured on the following shapes:

a

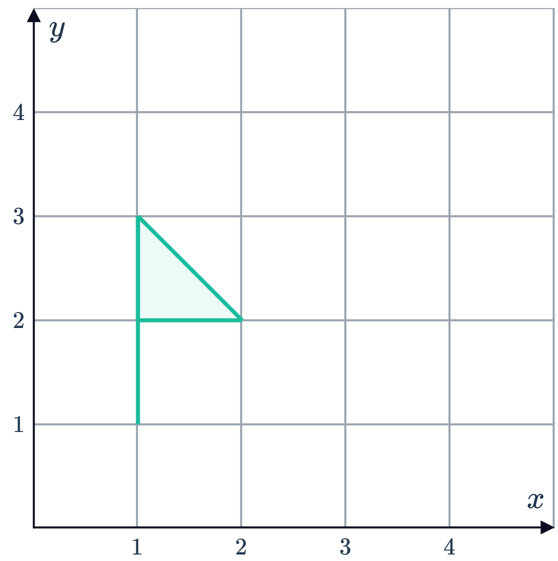
b



c

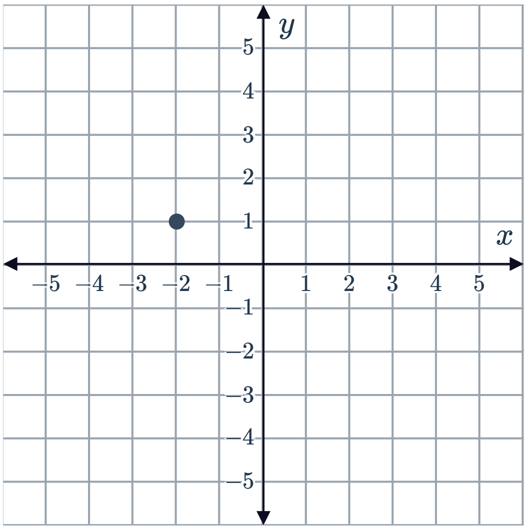


2 Translate the given flag 1 unit up and 2 units to the right on the number plane:

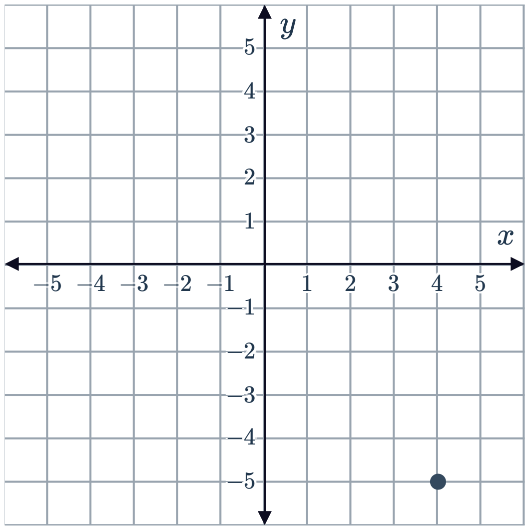


3 For each of the following points, plot the t[redacted]ated point using the given instruction:

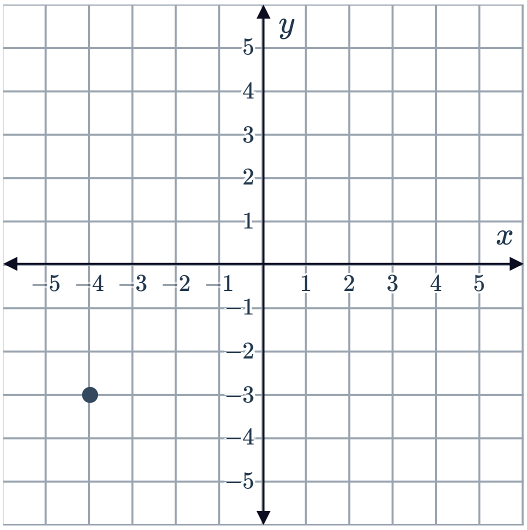
a Translate the point 3 units to the right.



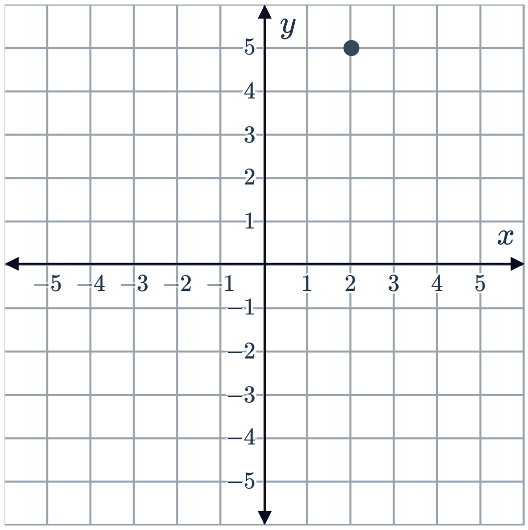
b Translate the point 4 units to the left.



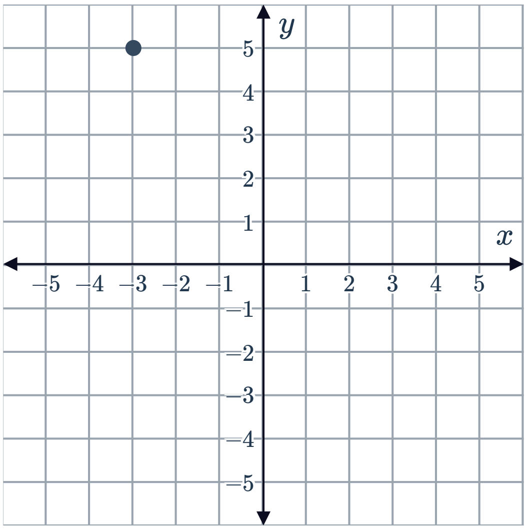
c Translate the point 7 units up.



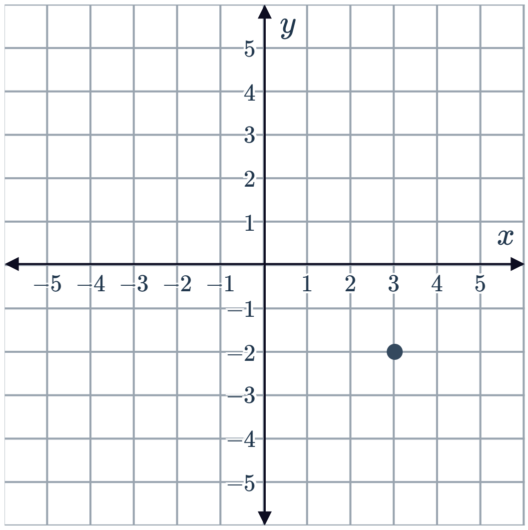
d Translate the point 4 units down.



e Translate the point 2 units to the right and 4 units down.

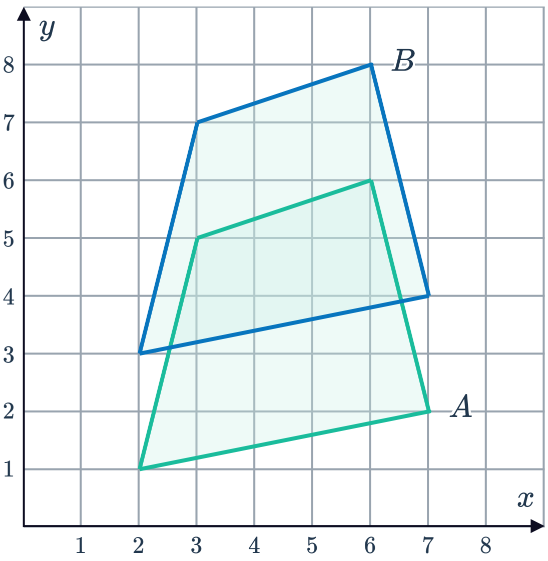


f Translate the point 5 units to the left and 6 units up.

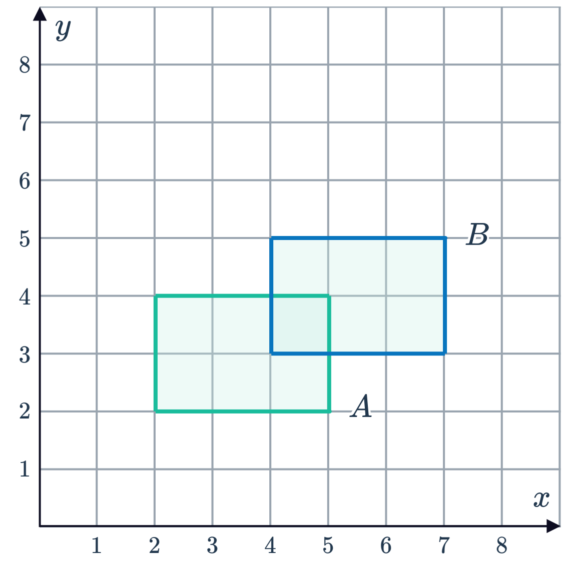


4 For each of the following, describe the tra[redacted]on from figure A to figure B:

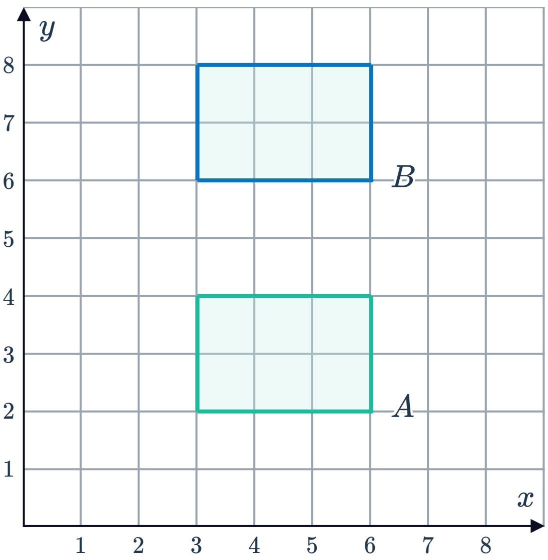
a



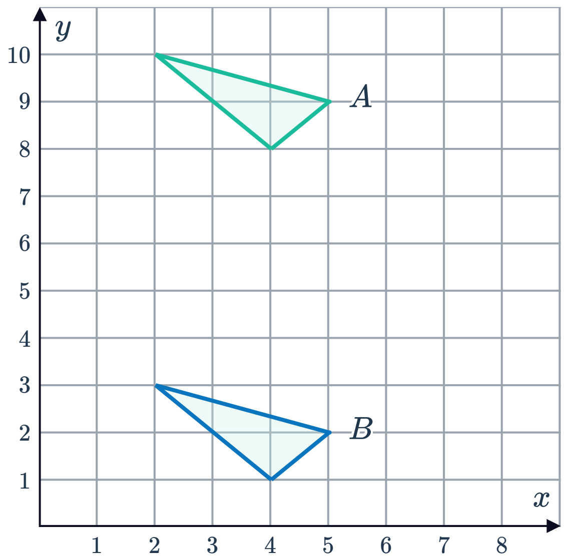
b



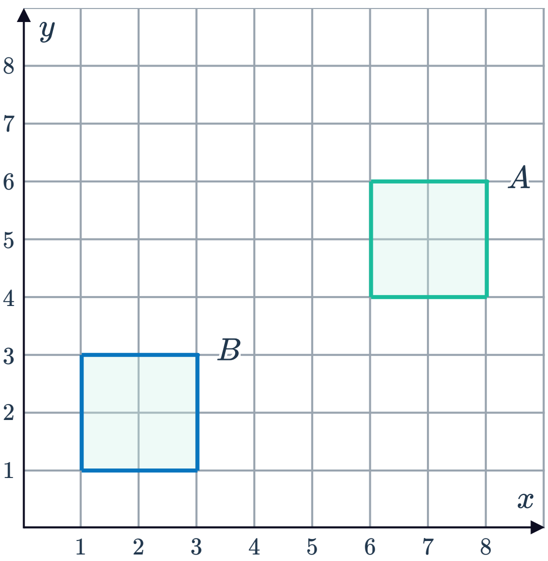
c



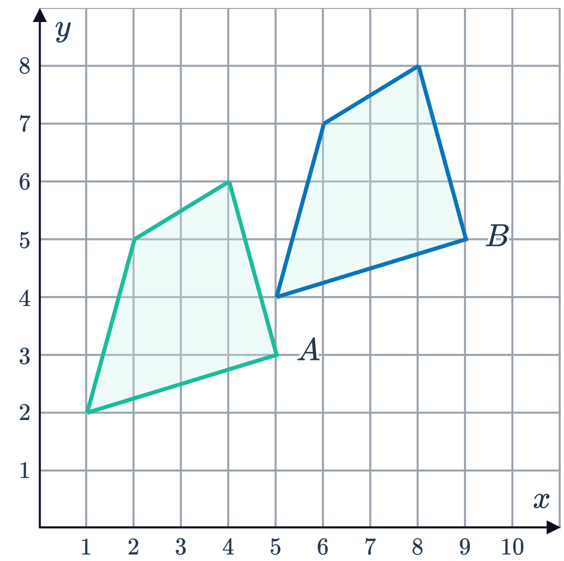
d



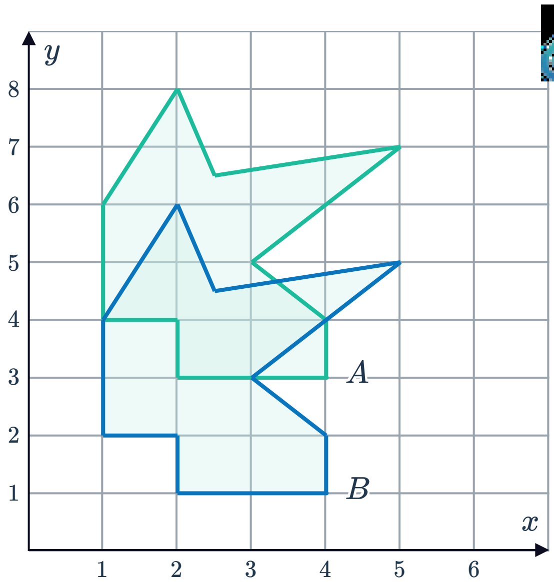
e



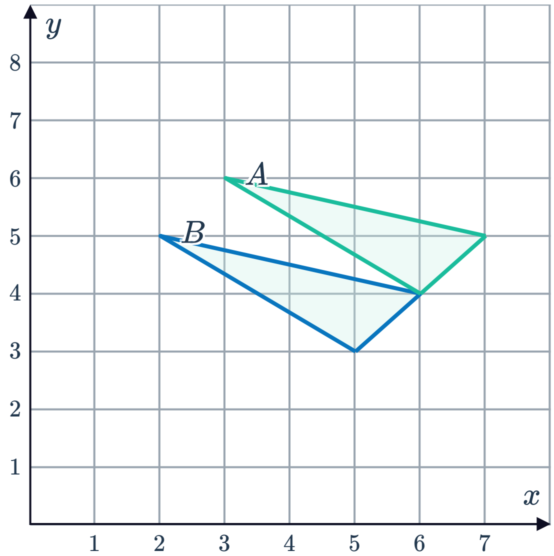
f



g

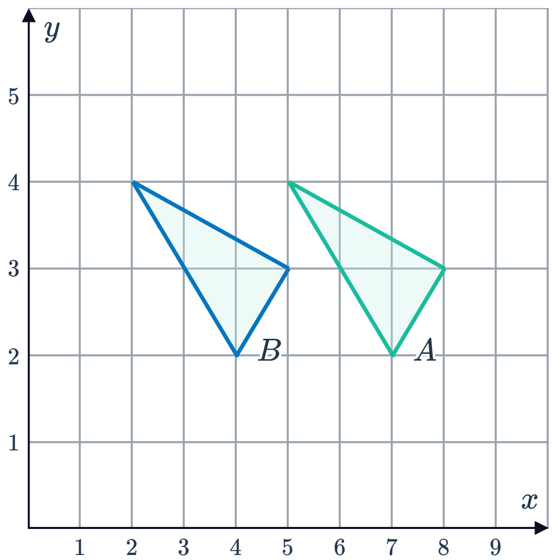


h

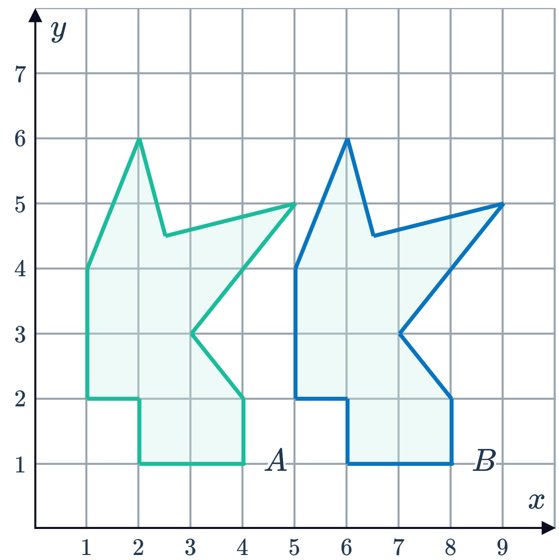


5 For each of the following, describe the translation from figure B to figure A:

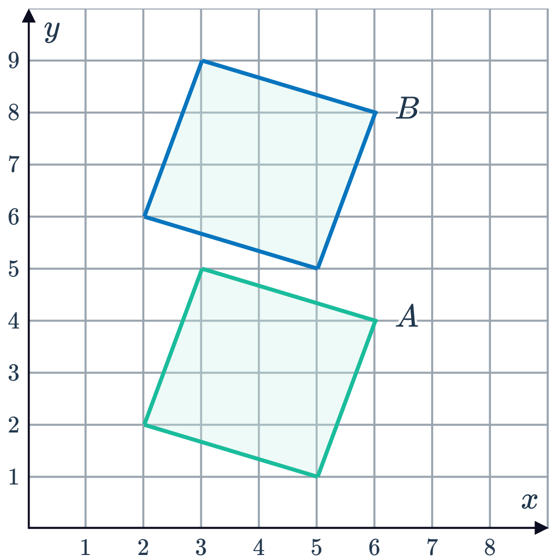
a



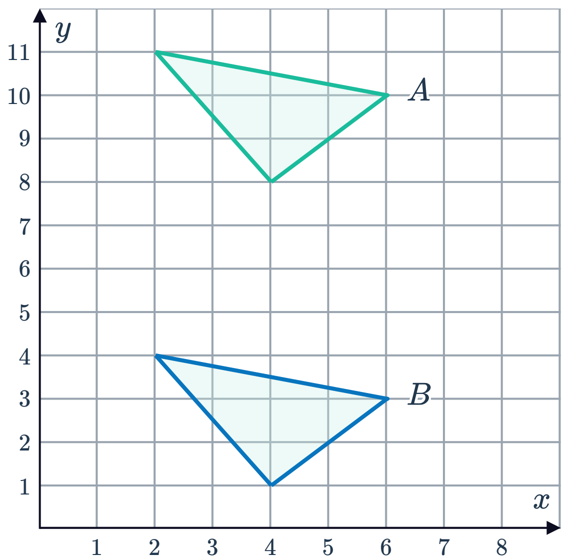
b



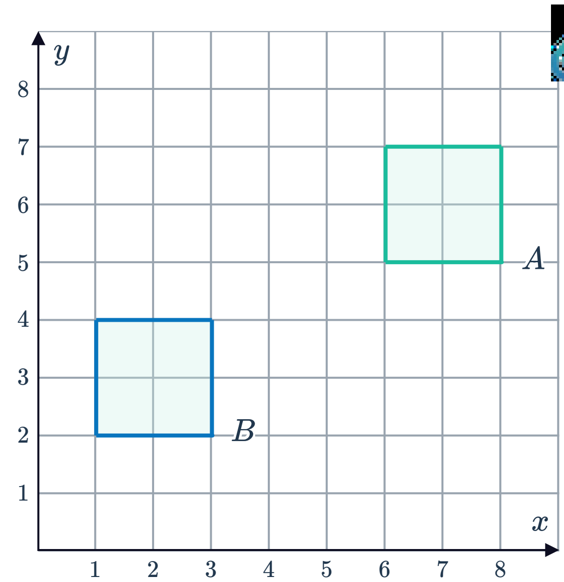
c



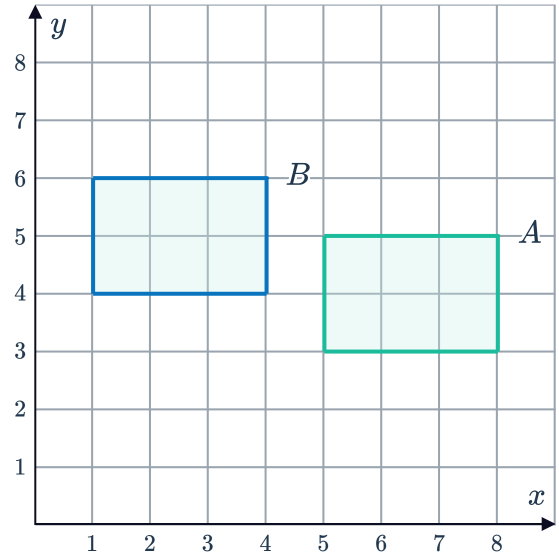
d



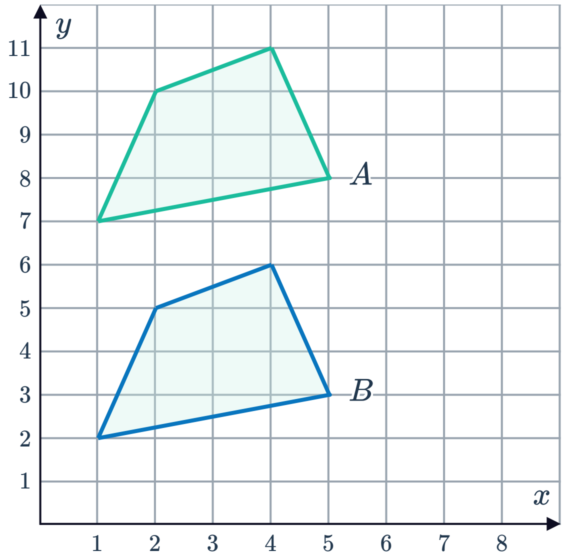
e



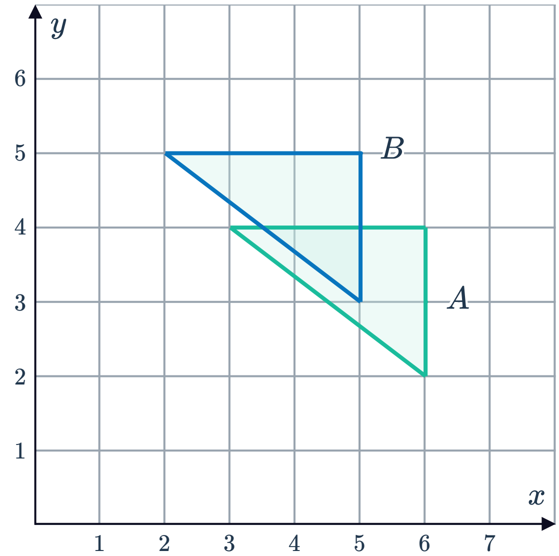
f



g

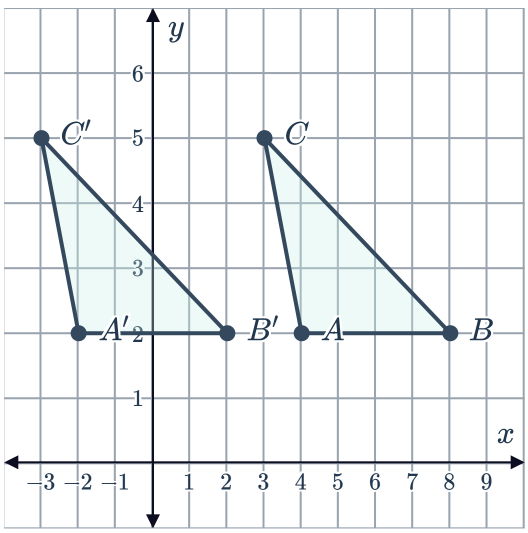


h

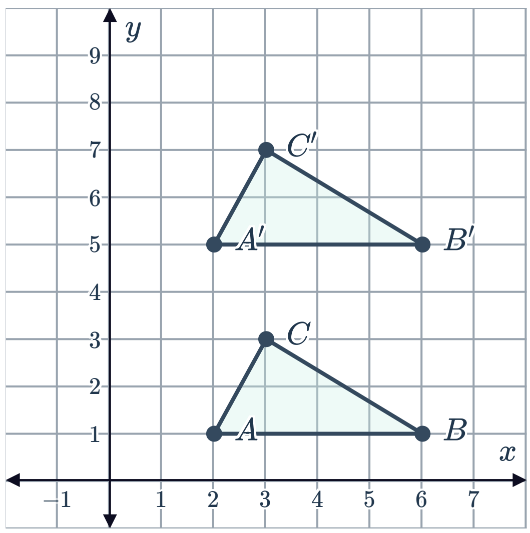


6 For each of the following, describe the translation of triangle ABC to triangle $A'B'C'$:

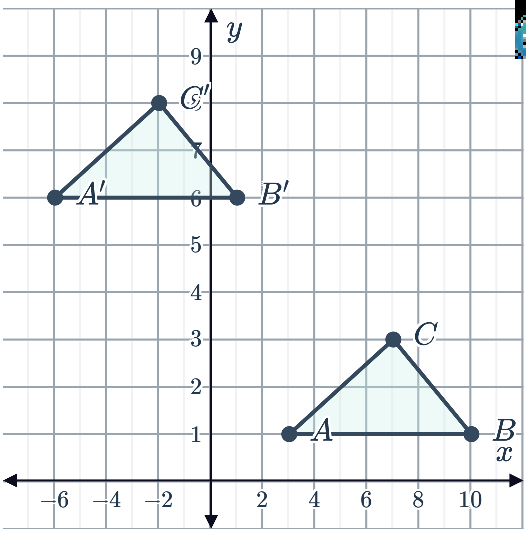
a



b

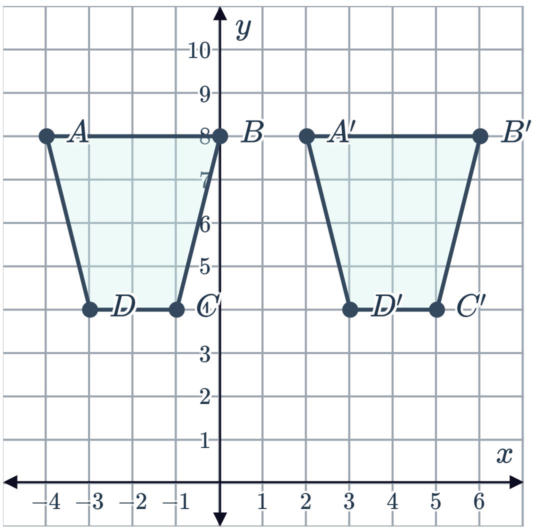


c

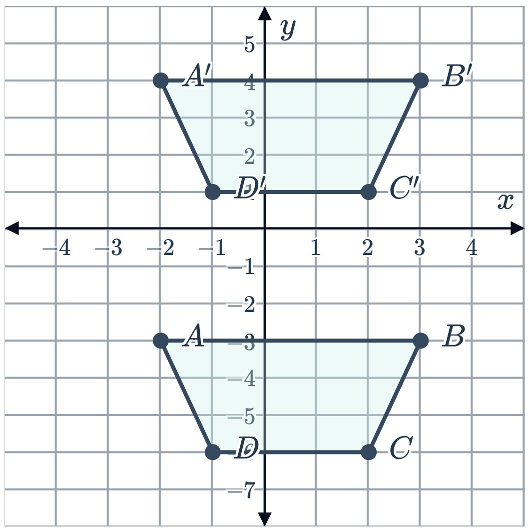


7 For each of the following, describe the translation of the trapezium $ABCD$ to the trapezium $A'B'C'D'$:

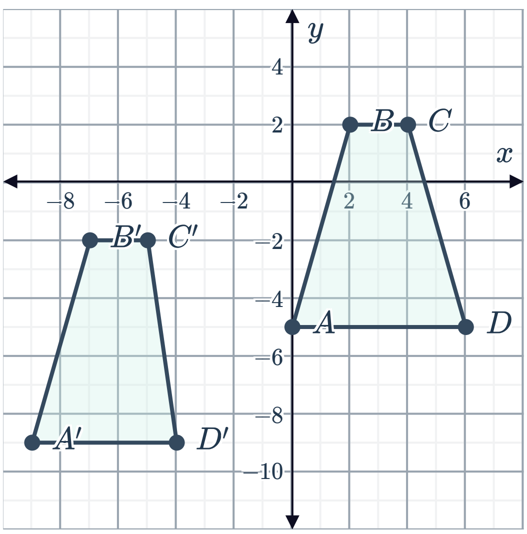
a



b

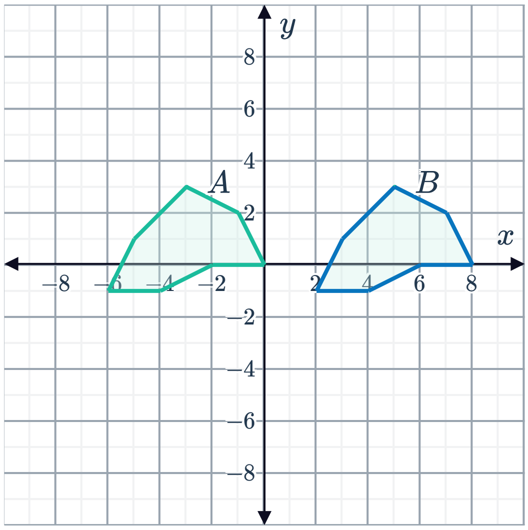


c

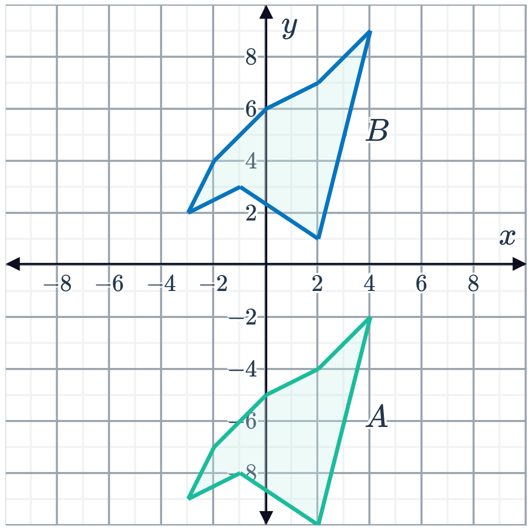


8 For each of the following, describe the tra[redacted]on of the polygon A to the polygon B:

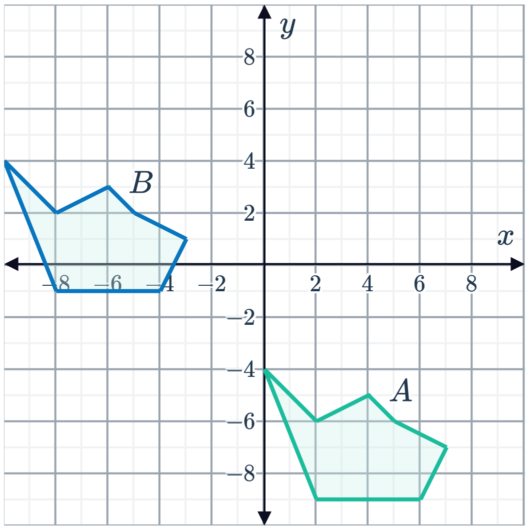
a



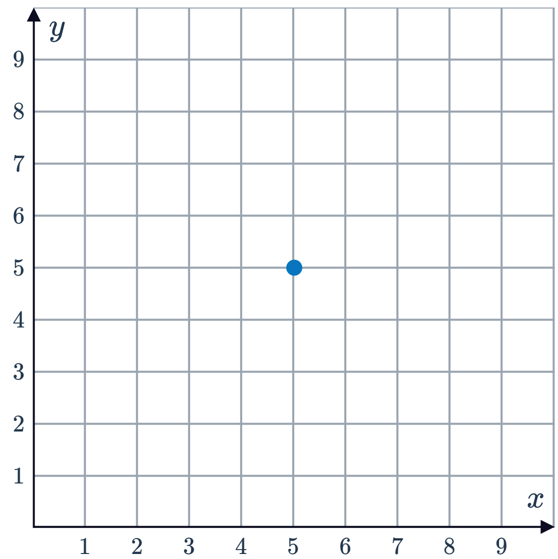
b



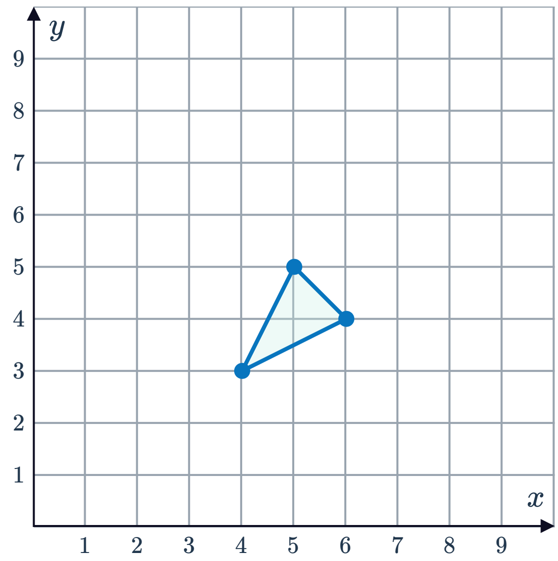
c



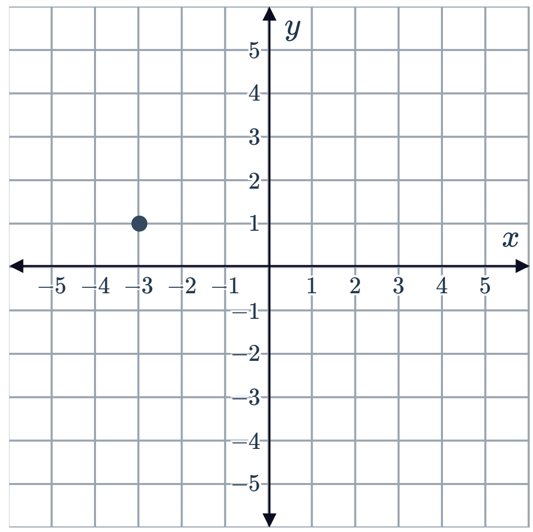
9 A point has been translated 2 units left. The translated point is shown on the right. Plot the starting position of this point before it was translated.



10 A triangle has been translated 3 units left. The translated triangle is shown on the right. Plot the starting position of the triangle before it was translated.



11 The point on the number plane is translated 4 units up. Write down the coordinates of its new position.

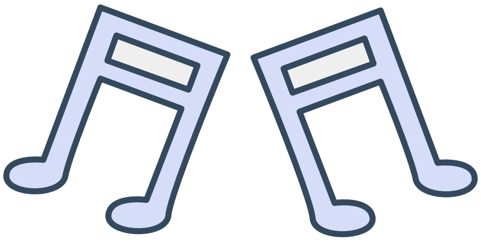


12 Point A is translated 3 units down and 4 units to the right, where it sits at point $(5, -2)$. Write down the coordinate of the original point A.

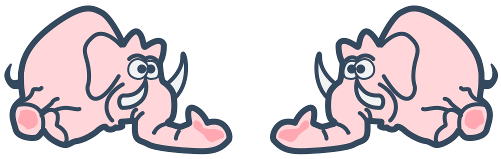
Reflections

13 State the type of transformation that has occured on the following shapes:

a

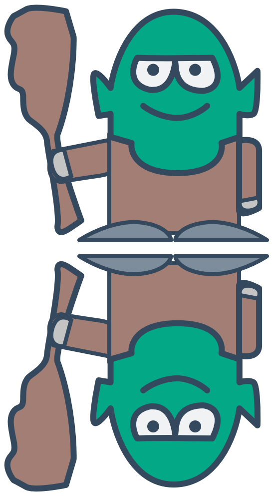


b



c

d

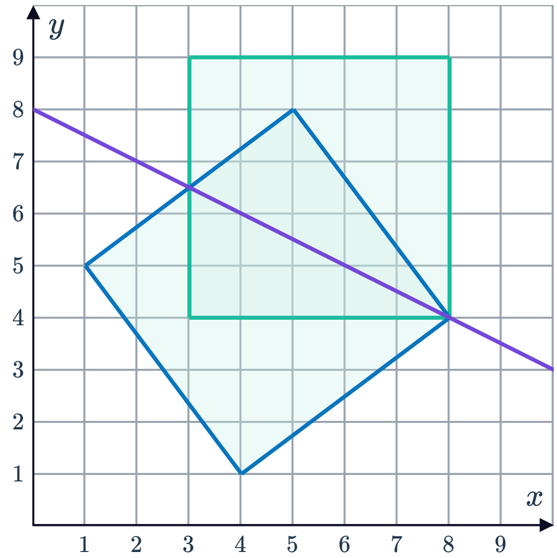


14 True or false:

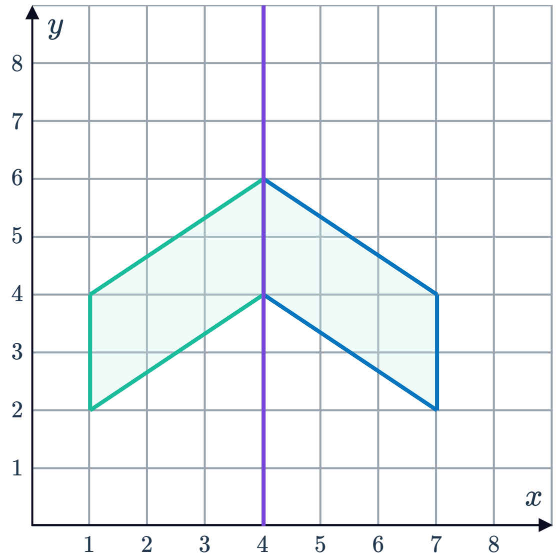
A shape is reflected across a line. If the reflected shape is then reflected across the same line again, it will return to the original shape.

15 Determine whether the following diagram[redacted]w a reflection across the given line:

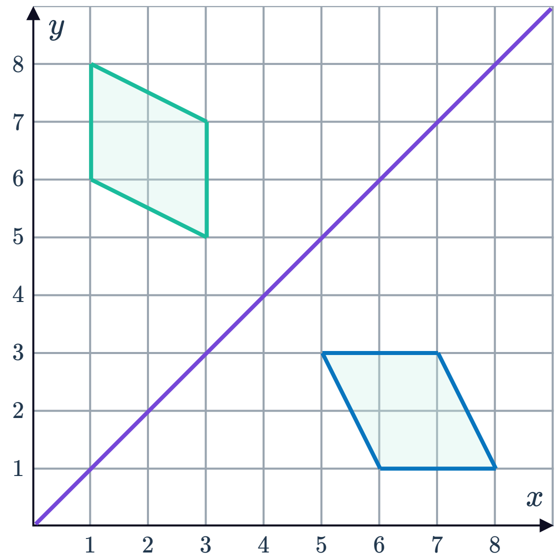
a



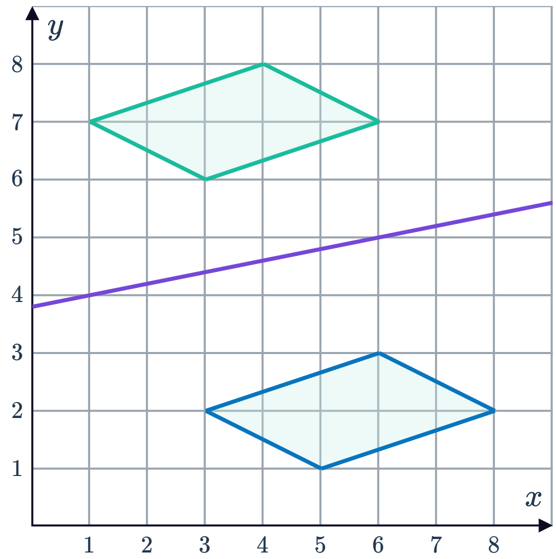
b



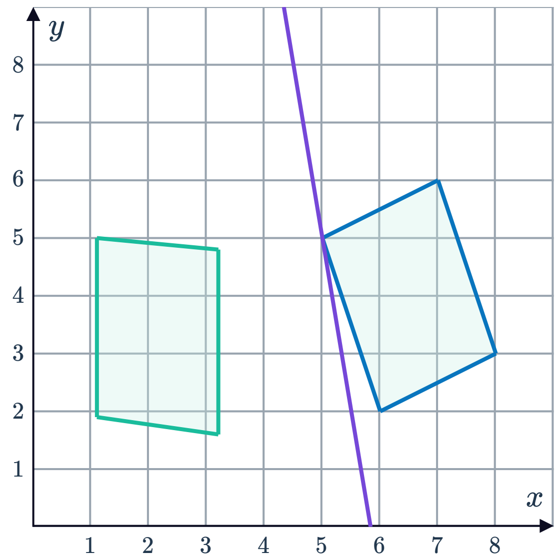
c



d

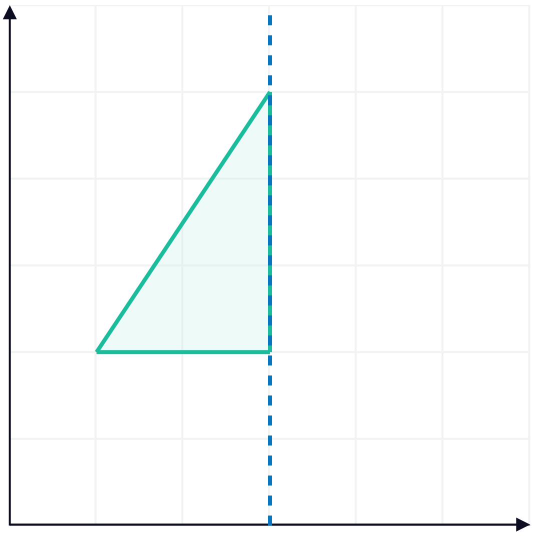


e

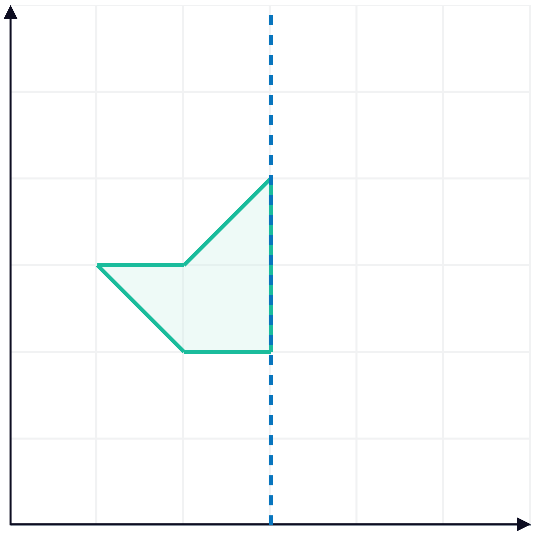


16 Reflect the following shapes about the giv[redacted]e:

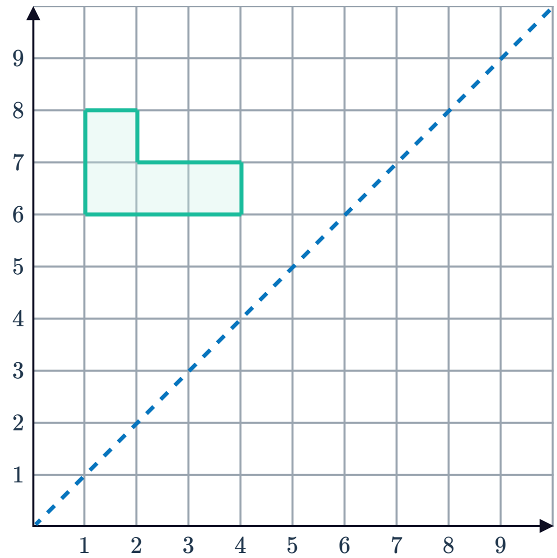
a



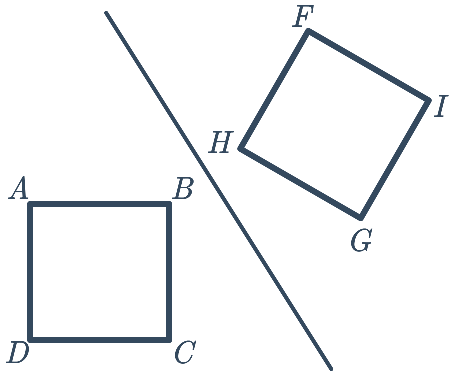
b



c



17 The square below is reflected about the given line. Which point would correspond to point A?



18 For each of the following:

- Plot the point A on a number plane.
- Plot point A', a reflection of point A across the indicated axis.

a $A\left(8,5\right)$; x-axis

b $A\left(-10,-6\right)$; x-axis

c $A\left(-5,-3\right)$; y-axis

d $A\left(8,-6\right)$; y-axis

19 Consider the point $A\left(5,6\right)$.

a Plot point A on a number plane.

b If point A is reflected across the y-axis to produce point B, what will be the coordinates of point B?

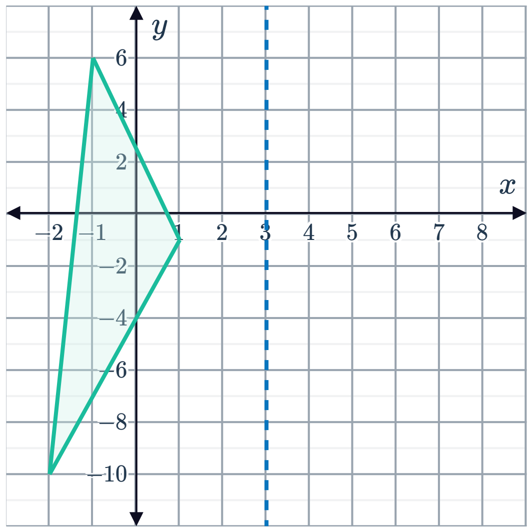
c What distance would the original point A need to be translated to overlap point B?

20 Consider the line segment AB, where the endpoints are $A\left(-9,-10\right)$ and $B\left(4,3\right)$.

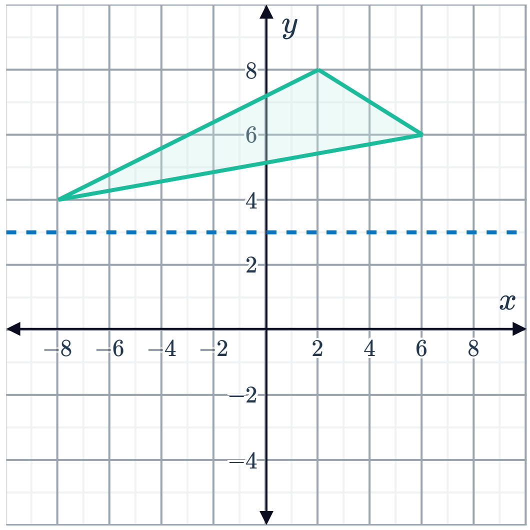
a Plot the line segment AB on a number plane.

b Plot the reflection of the line segment AB across the x-axis.

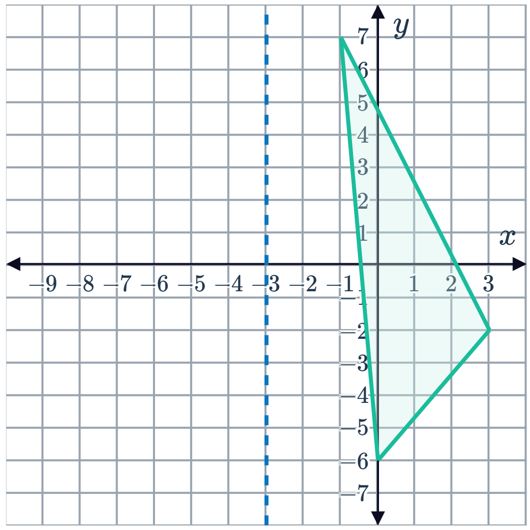
21 Plot the reflection of the given triangle across the line $x=3$:



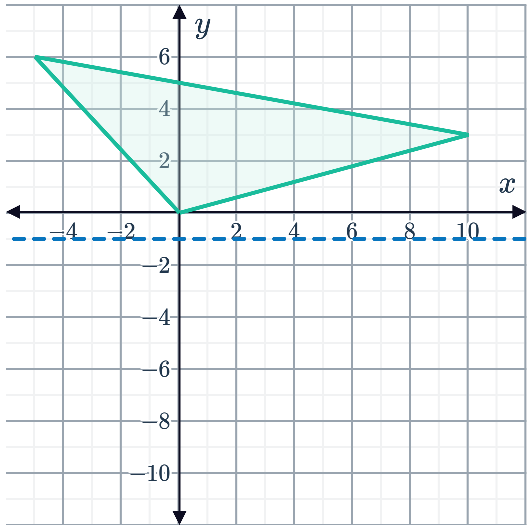
22 Plot the reflection of the given triangle across the line $y = 3$:



23 Plot the reflection of the given triangle across the line $x = -3$:

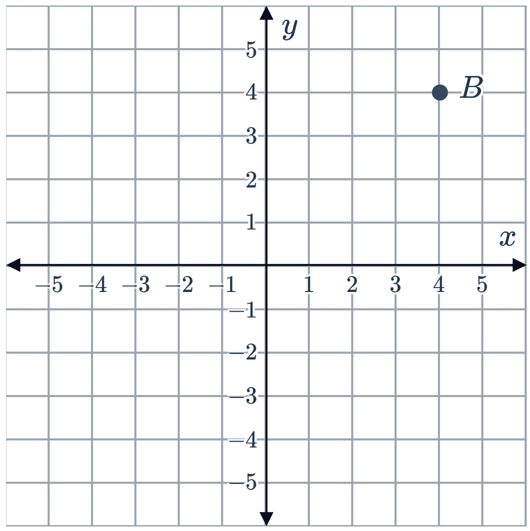


24 Plot the reflection of the given triangle across the line $y = -1$:



Rotations

25 Find the coordinates of point B when it is rotated 90° clockwise about the origin.



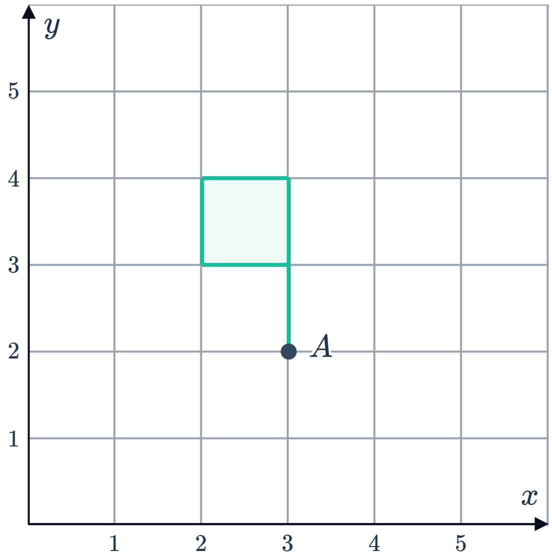
26 Consider the flag on the number plane shown:

Draw the flag after it has been rotated:

a 90° about point A.

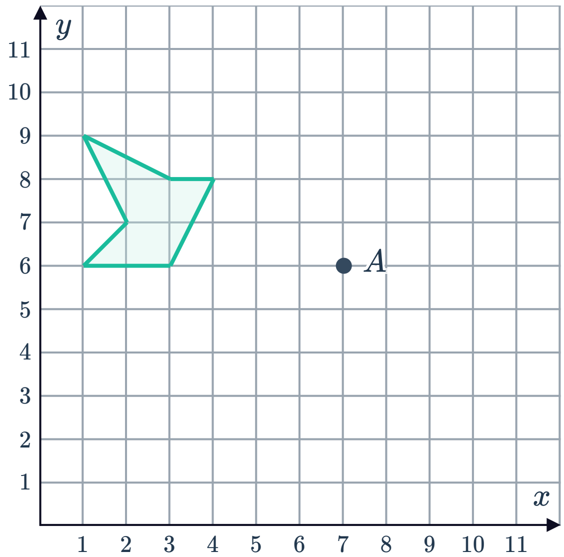
b 180° about point A.

c 270° about point A.



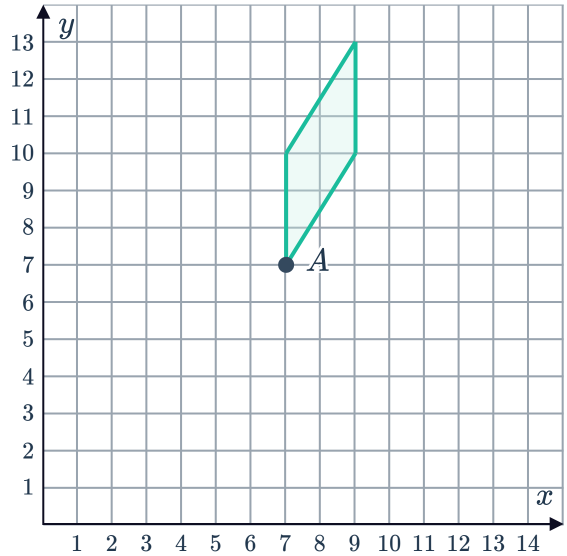
27 Consider the following shape:

Draw the shape after it has been rotated by 180° clockwise about point A.

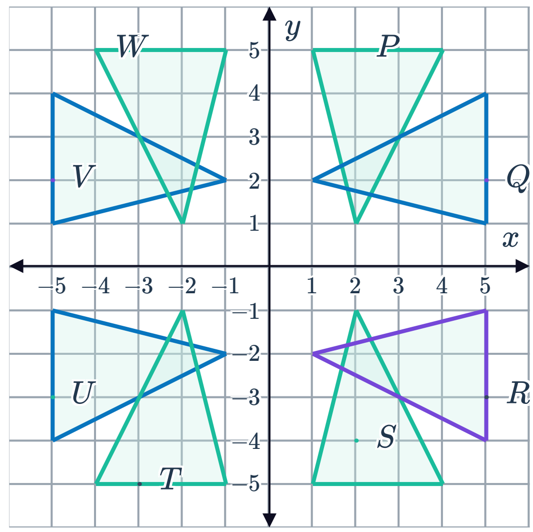


28 Consider the shape shown:

Draw the shape after it has been 135° anticlockwise rotated about point A.

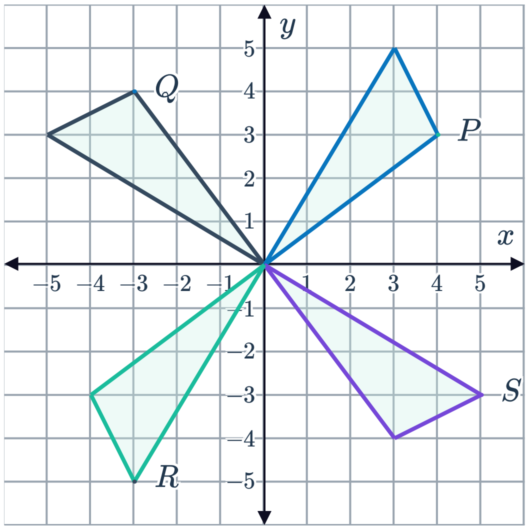


29 Which shape corresponds to a rotation of triangle R by 270° clockwise about the origin?

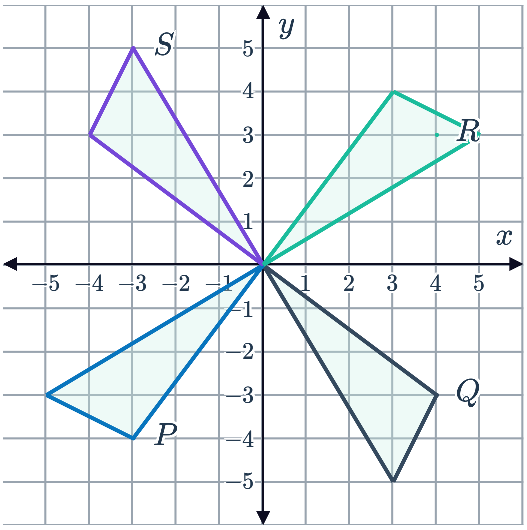


30 Suppose all four triangles in the image below are rotated 270° clockwise about the origin.

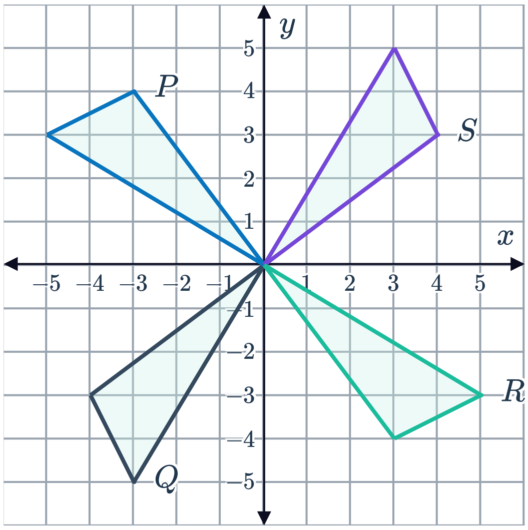
Which image would correspond to the result of this transformation?



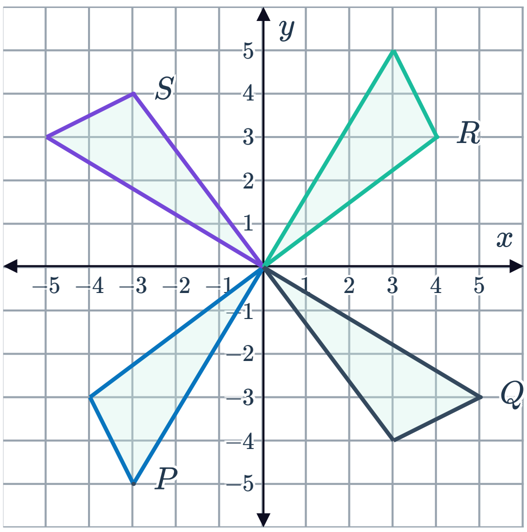
A



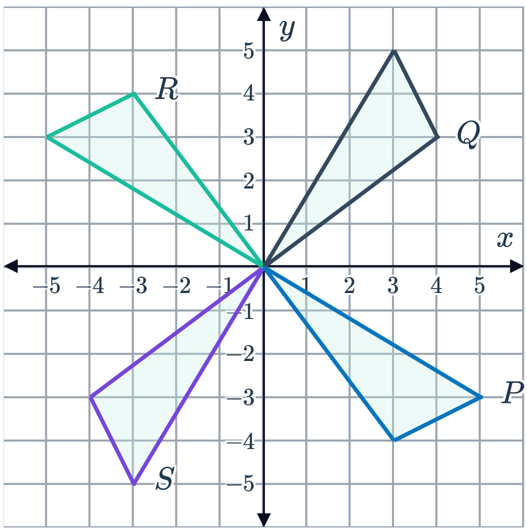
B



C



D

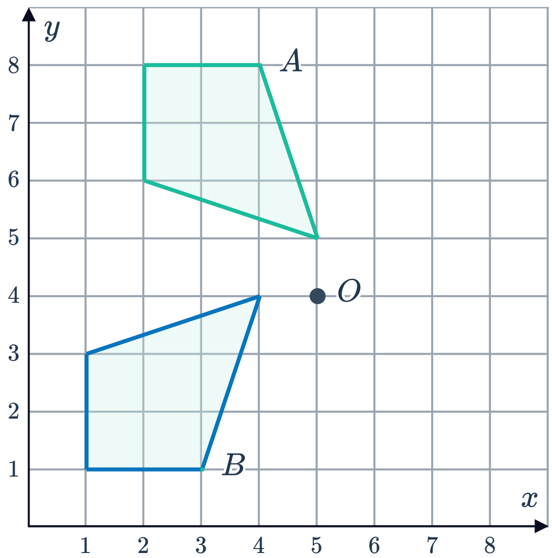


31 For each of the following:

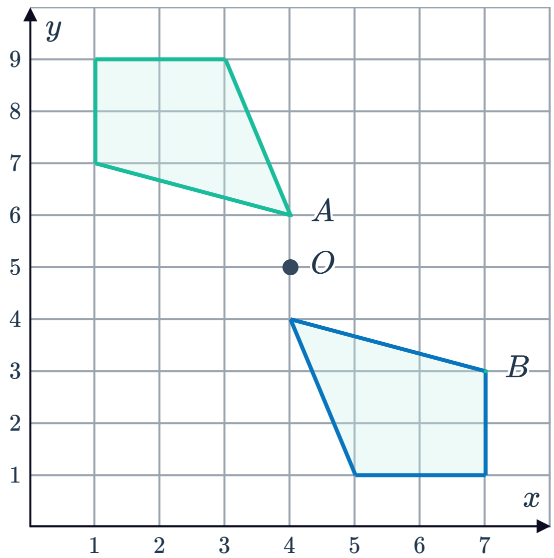
i Find the clockwise angle of rotation of shape A to shape B about point O.

ii Find the anticlockwise angle of rotation of shape A to shape B about point O.

a



b



32 Consider the following image:

a Which segment represents a clockwise rotation of segment PQ by 90° about point O?

b Which segment represents an anticlockwise rotation of segment XY by 270° about point O?